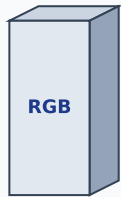


INPUT PREPROCESSING



3×256^2

Geometric Alignment

YuNet 5-Pt Landmarks

4-DoF LMEDS Transform

Scale Margin $s_f = 1.5 \times$

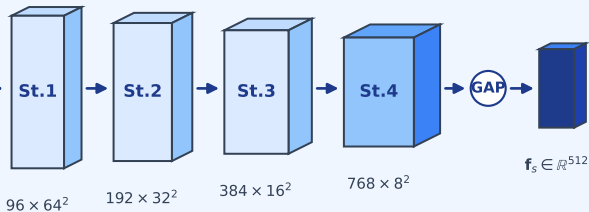
Boundary Windowing

Hann Taper Window

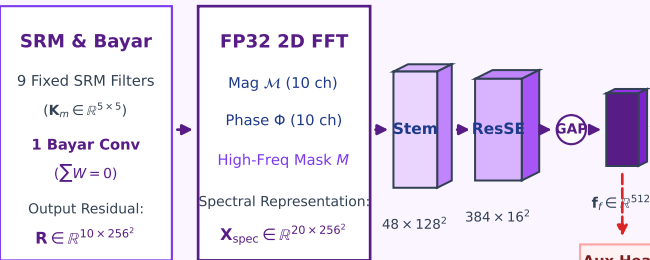
$H_{\text{taper}} = 12$ Pixels

Suppresses Spectral Seams

SPATIAL STREAM: ConvNeXt-Small Backbone (49.98M Params)



SPECTRAL STREAM: ResSE-Spectral Tower (2.99M Params)



Aux Head
 $\mathcal{L}_{\text{aux}}$ (BCE)

SNR-ADAPTIVE GATING

Noise Power Estimation

$$P_{\text{noise}} = \frac{1}{|\Omega_{\text{high}}|} \sum_{\mathbf{u} \in \Omega_{\text{high}}} \mathcal{M}^2(\mathbf{u})$$

Piecewise SNR Attenuation:

$$\gamma = \text{clip}\left(\frac{P_{\text{noise}}}{\varepsilon_1 - \varepsilon_0}, 0, 1\right)$$

$$\varepsilon_0 = 0.05, \quad \varepsilon_1 = 0.25$$

Complementary Fusion

Raw MLP Gate Logits:

$$\mathbf{g} = [g_s, g_f]^\top = \text{MLP}([\mathbf{f}_s \parallel \mathbf{f}_r])$$

Effective Gate Modulation:

$$\mathbf{g}_{\text{eff}} = [g_s, g_f \cdot \gamma]^\top$$

L1-Normalized Weights:

$$\tilde{\mathbf{g}} = \mathbf{g}_{\text{eff}} / \|\mathbf{g}_{\text{eff}}\|_1 = [\tilde{g}_s, \tilde{g}_f]^\top$$

Calibrated Multimodal Vector:

$$\mathbf{e}_t = [\tilde{g}_s \mathbf{f}_s \parallel \tilde{g}_f \mathbf{f}_f] \in \mathbb{R}^{1024}$$

SPATIOTEMPORAL HEAD & BAYESIAN TRIAGE

2-Layer Bidirectional GRU ($d_h = 256$)

First-Order Velocity Delta: $\Delta \mathbf{e}_t = \mathbf{e}_t - \mathbf{e}_{t-1}$

Augmented Frame Input: $[\mathbf{e}_t \parallel \Delta \mathbf{e}_t] \in \mathbb{R}^{2048}$

Dual Pooling: $\mathbf{c}_{\text{attn}} \oplus \mathbf{h}_{\text{max}} \rightarrow \mathbf{z}_{\text{video}} \in \mathbb{R}$

Captures Inter-Frame Temporal Coherence

Affine Platt Temperature Calibration

$$\hat{p} = \sigma(0.2783z_{\text{video}} + 0.4089), \quad T_{\text{eff}} = 3.5931$$

ECE Reduced: 0.1965 $\rightarrow$ **0.0994 (-49.4% Error Reduction)**

Guarantees Bayesian Risk Monotonicity

ZONE 1

Clearance

$$\hat{p} < 0.40$$

Cost = 0

ZONE 2

Manual Review

$$[0.40, 0.60)$$

$C_{\text{rev}} = 0.15$

ZONE 3

Interdiction

$$\hat{p} \geq 0.60$$

$C_{\text{flag}} = 1.0$